Sentiment Analysis Extension

https://github.com/aiyshwary/Video-based-and-audio-based-sentiment-analysis

Python | Hugging Face | NLP | RapidMiner | Audio Processing

OVERVIEW

A multimodal sentiment analysis project that independently extracts and analyses sentiment from both the video (visual) and audio streams of a recording, combining insights from two separate NLP and signal processing pipelines.

IMPACT

Built a dual-stream sentiment analysis pipeline that separates video and audio from a recording and independently classifies sentiment from each modality, enabling richer multimodal emotional understanding.

TECHNICAL WALKTHROUGH

Problem Statement

Understanding customer opinions and emotional tone from text data is crucial for decision-making in domains like customer feedback, reviews, and social media.

Approach

Built a sentiment analysis pipeline using

i) NLP preprocessing

ii) Pre-trained language models

Supports classification into

i) Positive

ii) Negative

iii) Neutral sentiments

iv) Methodology

v) Text data is cleaned and vectorized

Models are trained or fine-tuned on labeled sentiment datasets

Predictions include

i) Sentiment label

ii) Confidence score

iii) Applications

iv) Customer feedback analysis

v) Product review monitoring

vi) Social media sentiment tracking

vii) Deployment

Entire system deployed through AI-Hub using RapidMiner

Images and data stored in AWS S3

Inference handled via automated workflows

Supports

i) Real-time predictions

ii) Error handling

iii) Scalable deployment

iv) Conclusion

This project delivers a complete AI ecosystem for

i) Image Classification

ii) Object Detection

iii) Sentiment Analysis

The key strength lies in its flexibility, modularity, and deployability, allowing

users to

i) Choose models based on requirements

ii) Fine-tune on custom datasets

iii) Deploy seamlessly in production environments

It bridges the gap between research-level models and real-world industrial applications.

KEY DETAILS

1. Extracts audio track from video files and applies speech-to-text followed by NLP-based sentiment classification.

2. Processes video frames independently to infer visual sentiment cues from facial expressions and scene context.

3. Outputs positive, neutral, or negative sentiment labels with confidence scores per modality.

4. Designed for use cases including customer feedback analysis, social media monitoring, and presentation review.

5. Implemented using Hugging Face pretrained models and RapidMiner for pipeline orchestration.